DATA SCIENCE REPORT

# PRODUCT TAXONOMY INTELLIGENCE: NLP-BASED PRODUCT CLASSIFICATION & EXPLAINABILITY

## Character-Level Feature Engineering, Machine Learning & Classification Performance



**TOOLS USED**

Python, Pandas, Numpy, Matplotlib, Plotly & ML libraries

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# Overview & Business Objective

# 1.1 PROJECT OVERVIEW

E-commerce platforms and product comparison services manage large catalogs containing thousands of product listings from different merchants. These listings often vary in naming conventions, technical specifications, brand references, and product descriptions.

Manually categorizing such products can be time-consuming and inconsistent, particularly when similar products are listed under different names.

The Product Taxonomy Intelligence project applies Natural Language Processing (NLP) and machine learning to automatically classify products into predefined categories using their titles.

The project covers data exploration, text preprocessing, TF-IDF feature engineering, classical machine learning, TextCNN experimentation, model evaluation, error analysis, and explainability.

# 1.2 PROBLEM STATEMENT

Product titles frequently contain abbreviations, technical specifications, model identifiers, and overlapping terminology, making accurate classification challenging.

The key problem addressed in this project is:

***How can machine learning classify products into predefined categories using product titles while maintaining strong performance across different product types?***

The project formulates this as a 10-class supervised text classification problem, with *Category Label* as the target variable.

# 1.3 PROJECT OBJECTIVES

The major objectives of the project are:

- Analyze product listings and identify category-specific text patterns.
- Prepare product titles for NLP-based classification.
- Develop and compare word-level and character-level TF-IDF representations.
- Train and optimize multiple machine learning classifiers.
- Evaluate a TextCNN architecture as a deep learning alternative.
- Prevent information leakage through group-aware data splitting and cross-validation.
- Assess classification performance and investigate prediction errors.
- Interpret model decisions using character-level feature contributions.

# 1.4 PROJECT SCOPE

The project follows a structured workflow covering data analysis, NLP-based feature engineering, machine learning classification, model explainability, and business interpretation to identify product naming patterns, evaluate classification performance, understand model decisions, and explore opportunities for automated product categorization.

01

## Data Analysis

- Dataset inspection
- Data quality assessment
- Category distribution analysis
- Merchant and cluster analysis

02

## NLP & Classification

- Text preprocessing
- TF-IDF feature extraction
- Group-aware validation
- Machine learning & TextCNN modeling

03

## Model Explainability

- Character feature importance
- Misclassification analysis
- Decision margin analysis

04

## Business Interpretation

- Automated product categorization
- Product catalog consistency
- Ambiguous listing identification
- Classification limitations
- Recommendations for improvement

# Data Understanding & Exploratory Data Analysis

# 2.1 DATA OVERVIEW

The dataset provides a comprehensive view of product listings across multiple merchants and categories, covering product titles, identifiers, category labels, and product clustering information. These variables provide the foundation for analyzing product naming patterns, identifying category-specific terminology, and developing NLP-based machine learning models for accurate product classification.

***Data Source: Kaggle***

## Dataset Size

**35,311**
Records

**16**
Original Features

## Data Diversity

**10**
Categories

**306**
Merchamts

**12,849**
Clusters



# DATA OVERVIEW

**Below is a detailed description of the feature set:**

| Dataset Features | Type | Feature Description |
|---|---|---|
| Product ID | Categorical (Identifier) | Unique listing identifier |
| Product Title | Text (Unstructured) | Textual product name |
| Merchant ID | Categorical (Identifier) | Identifier of the merchant |
| Cluster ID | Categorical (Identifier) | Product grouping identifier |
| Cluster Label | Text (Categorical Label) | Product grouping description |
| Category ID | Categorical (Nominal) | Numerical product category identifier |
| Category Label | Categorical (Nominal) | Product category name |

# 2.2 TARGET VARIABLE

The target variable selected for prediction is:

*Demand*

Category Label represents the product category that the model attempts to predict based on the textual information contained in each product title.

The target has:

**10**
Unique Categories

**2,212**
Smallest Category Count

**5,501**
Largest Category Count

**15.58%**
Largest Category Share

**6.26%**
Smallest Category Share

The distribution is moderately imbalanced, with Fridge Freezers representing the largest category and Freezers the smallest.

# 2.3 DATA QUALITY & PREPARATION OVERVIEW

Before modeling, the dataset was examined for:

- Data types
- Missing values
- Duplicate observations
- Categorical variables
- Numerical variables
- Temporal information
- Potential outliers
- Feature relationships

The assessment identified:

- 35,311 observations and seven variables.
- No missing values across the original columns.
- No fully duplicated rows.
- 30,993 unique product titles, indicating repeated titles across listings.

Following data preparation, 35,310 observations were retained for modeling.

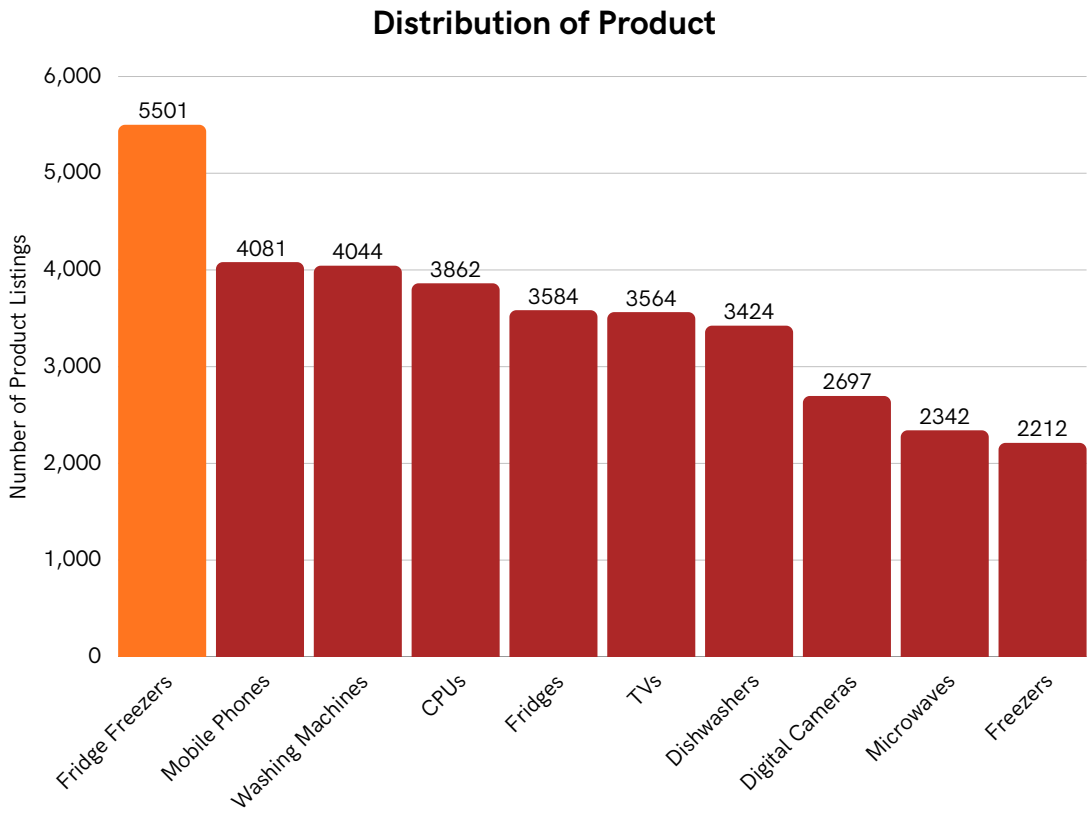
Repeated product titles and product clusters were important considerations when designing the train-test split to reduce information leakage.

# 2.4 Exploratory Data Analysis

# Product categories are moderately imbalanced, with Fridge Freezers representing the largest share of listings.

**15.58%**

Fridge Freezer Listings




## Key observations

- Fridge Freezers account for approximately 15.58% of listings, representing the largest category.
- Freezers account for approximately 6.26%, making them the least represented category. The remaining categories have relatively moderate differences in representation.

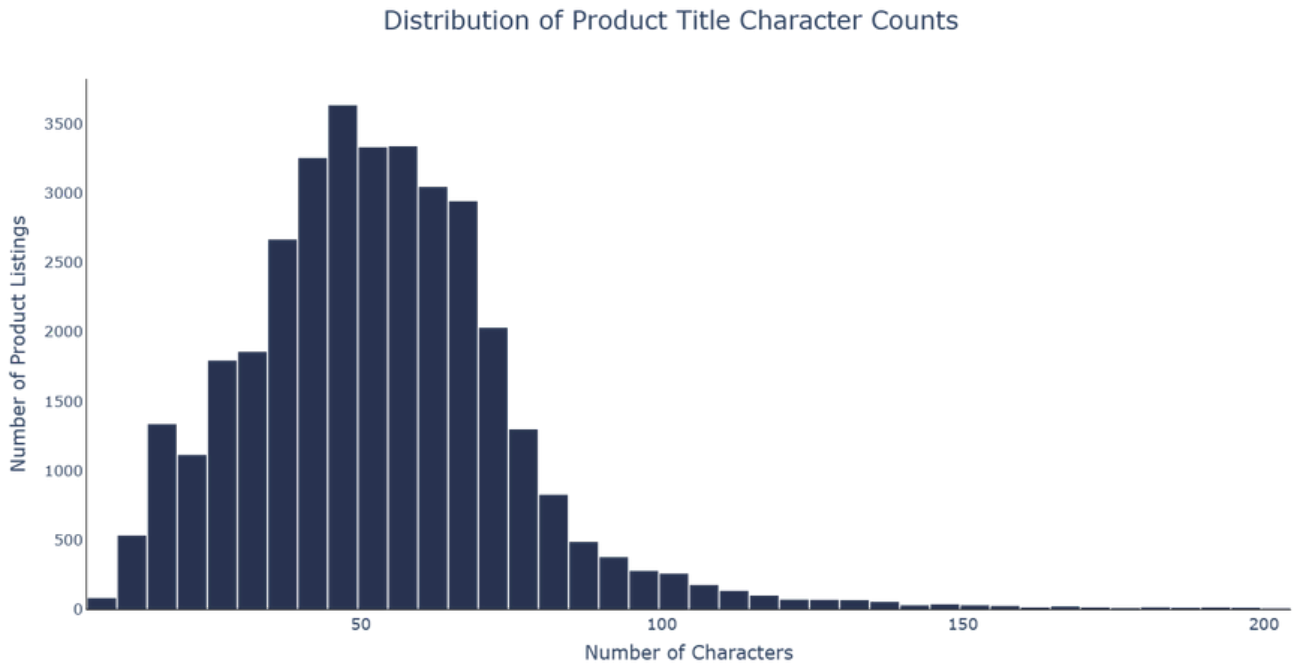
## Business Insights

- Category-level evaluation is necessary to identify performance differences that overall accuracy may conceal.
- Macro-averaged metrics provide a useful assessment of classification performance across categories.

# Product titles vary considerably in length, averaging approximately 53 characters and nine words per listing.

**53**

Average Title Length




## Key observations

- Product titles contain approximately 53 characters and 8.67 words on average.
- Title lengths range from 5 to 200 characters.
- Product descriptions vary considerably in the amount of information provided.

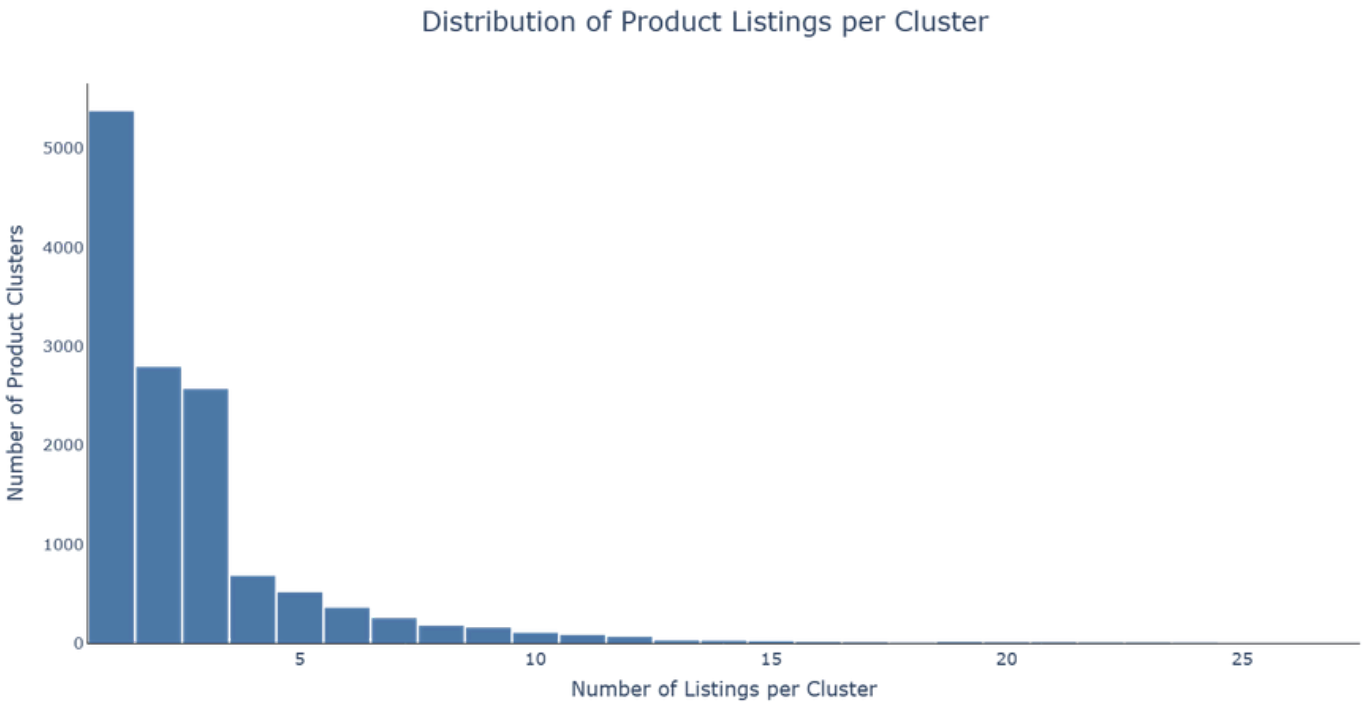
## Business Insights

- Short titles may provide limited evidence for classification.
- Character-level representations can capture useful patterns within abbreviated names and model identifiers.

# Most product clusters contain multiple listings, indicating repeated representations of underlying products.

**59.39%**

Multi-Listing Cluster Share




## Key observations

- The dataset contains 13,233 unique product clusters. Of these, 7,859 clusters contain multiple listings.Individual clusters contain up to 27 listings.

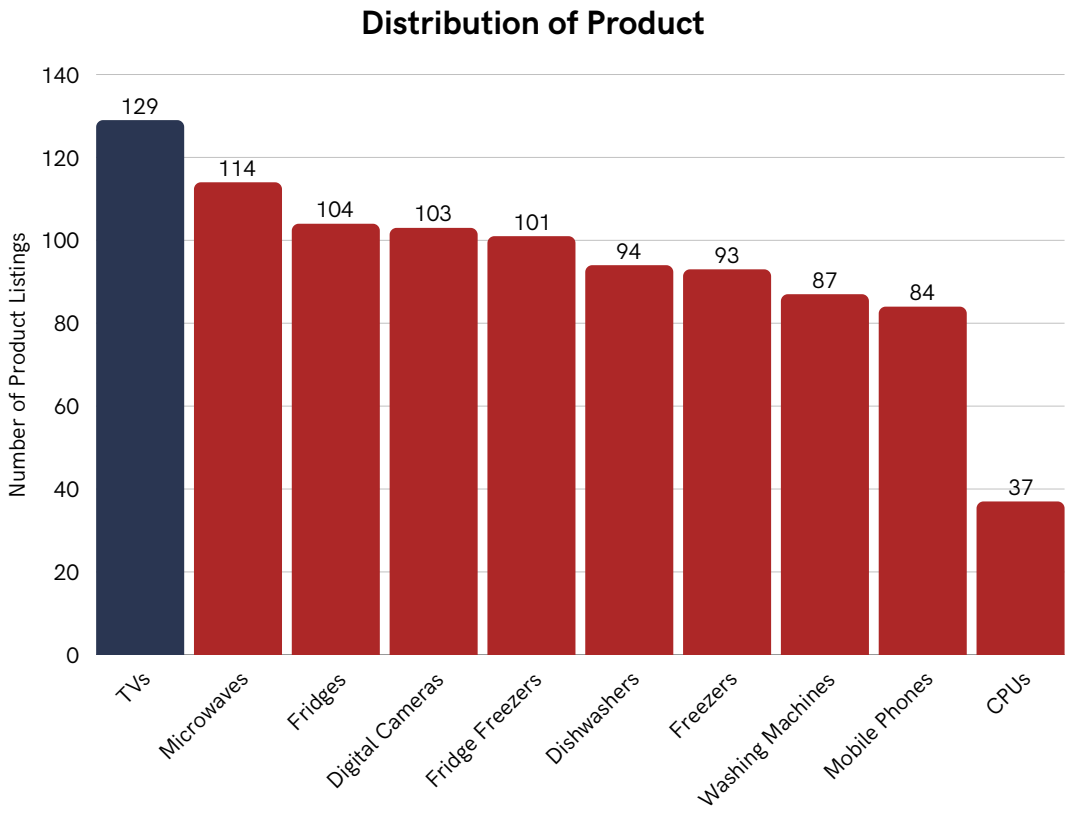
## Business Insights

- Multiple listings may refer to the same underlying product.
- Product grouping information must be considered during validation to avoid overly optimistic performance estimates.

# Product listings are unevenly distributed across merchants, with the largest merchant contributing substantially more listings.

**2,547**

Largest Merchant Listings




## Key observations

- The dataset contains 306 unique merchants. Merchant representation varies substantially, with the largest merchant contributing 2,547 listings.
- Some merchants specialize in a single category, while others list products across multiple categories.

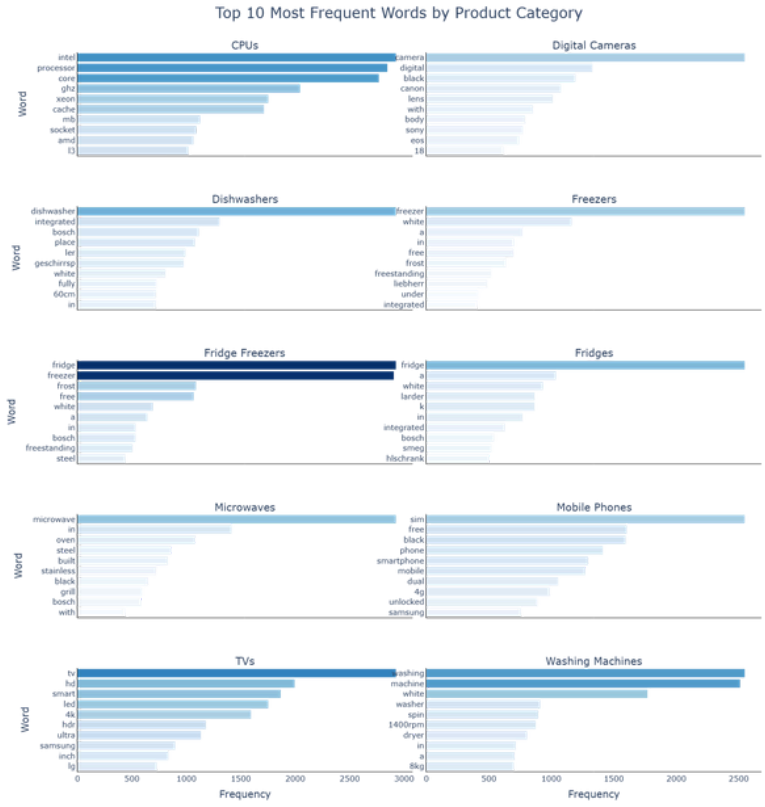
## Business Insights

- Product naming patterns may vary between merchants.
- A robust classification system should learn transferable product terminology rather than depend exclusively on merchant-specific patterns.

# Product categories exhibit distinctive terminology, although related appliance categories share overlapping keywords.

**10**

Product Categories Analyzed




## Key observations

- Product categories exhibit recognizable terminology associated with their functions and technical specifications.
- Common terms include fridge, freezer, intel, processor, washing, and smart.
- Related categories share terminology, particularly Fridges, Freezers, and Fridge Freezers.

## Business Insights

- Text-based classification is feasible because product titles contain meaningful category information.
- Overlapping terminology may introduce ambiguity between closely related categories.